

# Brooke Costin

Winthrop, MA | 617-899-2026 | brookecostin18@gmail.com | [LinkedIn](#) | [Portfolio](#)

## EDUCATION

**University of Rhode Island**

*Bachelor of Science in Biomedical Engineering*

*Minor in Mathematics*

**South Kingston, RI**

**Graduated: May 2025**

## TECHNICAL SKILLS

**CAD & Design:** SolidWorks | Shapr3D | Onshape | Fusion 360

**Programming:** Python | MATLAB | C++ | HTML/CSS | JavaScript

**Additive Manufacturing:** FFF/FDM 3D Printing | Bambu Lab | Raise3D | Bambu Studio

**Hardware:** Arduino | ESP32 | FDC1004 | Soldering | Oscilloscope

## ENGINEERING PROJECTS

**InSOLE 3D Printed Project** | Biomedical Engineering Capstone Design

- Designed and prototyped a fully 3D-printed capacitive sensing insole for real-time gait analysis and athletic performance monitoring.
- Optimized FFF printing parameters for flexible and conductive materials after encountering printer compatibility and print-quality issues.
- Iterated sensor geometry and material selection through testing, ultimately integrating Multi3D Electrifi sensors into a flexible TPU insole.

**Assembling an ECG Lab** | Biomeasurements Laboratory

- Constructing and experimenting with embedded systems utilizing microcontrollers, assembly, and C languages. Involves building an electrocardiogram (ECG) amplifier, configuring timers and interrupts for embedded signal processing, and implementing QRS detection and a heart rate meter.

**Digital Design 3D Printing Project** | Biomedical Engineering

- Designed an eight-component articulated LEGO figure in SolidWorks and manufactured the assembly using FFF 3D printing.

## EXPERIENCES

**Department of Biological Sciences**

**2023-2024**

*Teaching Assistant Anatomy and Physiology II*

- Facilitated learning of advanced human body systems, including cardiovascular, respiratory, and endocrine systems, and contributed to course content creation.

**Department of Biological Sciences**

**2023-2024**

*Laboratory Assistant*

- Maintained laboratory equipment and prepared wet labs for upcoming dissections. Collaborated well with peers in laboratory settings.
- Adept in specimen preparation, safety protocols, and equipment maintenance.

## WORK EXPERIENCE

**Belle Isle Seafood** | Winthrop MA

**2017-Present**

*Waitress and Bartender*

- Demonstrated strong time management skills by balancing 40+ hours of work and summer classes.
- Displayed organization skills while helping the owner order supplies daily to keep the restaurant stocked for the next day.